

Windows Forensic Acquisition and Artefact Analysis

Acquisition, integrity verification and artefact analysis

01

E01

FTK Imager
Autopsy 4.23.1
MFTECmd · PECmd ·
SBECmd

The investigation followed a defensible evidence workflow

1

Acquire

Create a forensic E01 image from the physical Windows drive.

2

Verify

Confirm the stored and computed hashes match.

3

Process

Load the image into Autopsy and run targeted ingest modules.

4

Interpret

Review partitions, OS details, Prefetch, MFT and event evidence.

Outcome: a repeatable process that preserves evidence integrity and links findings to source artefacts.

Each tool answered a different forensic question

FTK Imager

Can the physical source be captured and verified without altering the original evidence?

Acquisition + hashing

Autopsy

What partitions, system details and user-activity artefacts exist in the E01 image?

Triage + artefact review

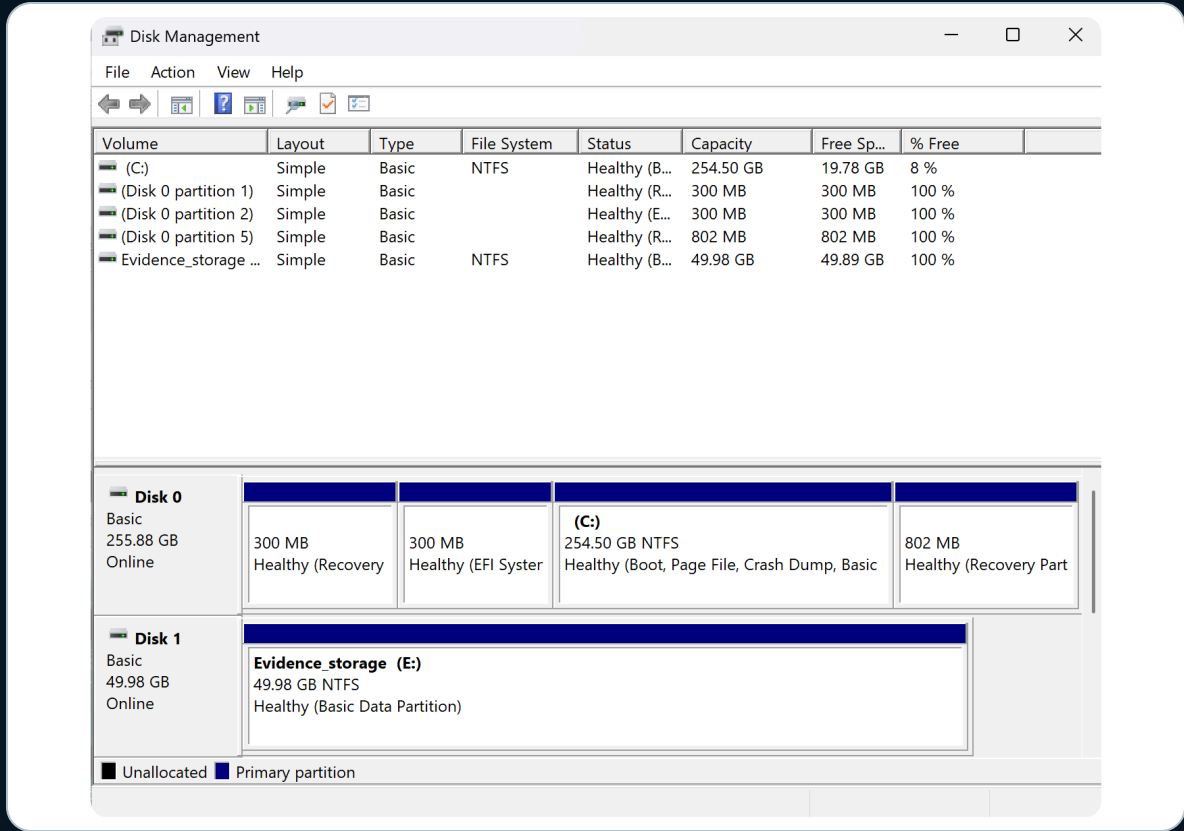
Zimmerman tools

What do NTFS, Prefetch and registry-derived records reveal at a lower level?

Targeted parsing

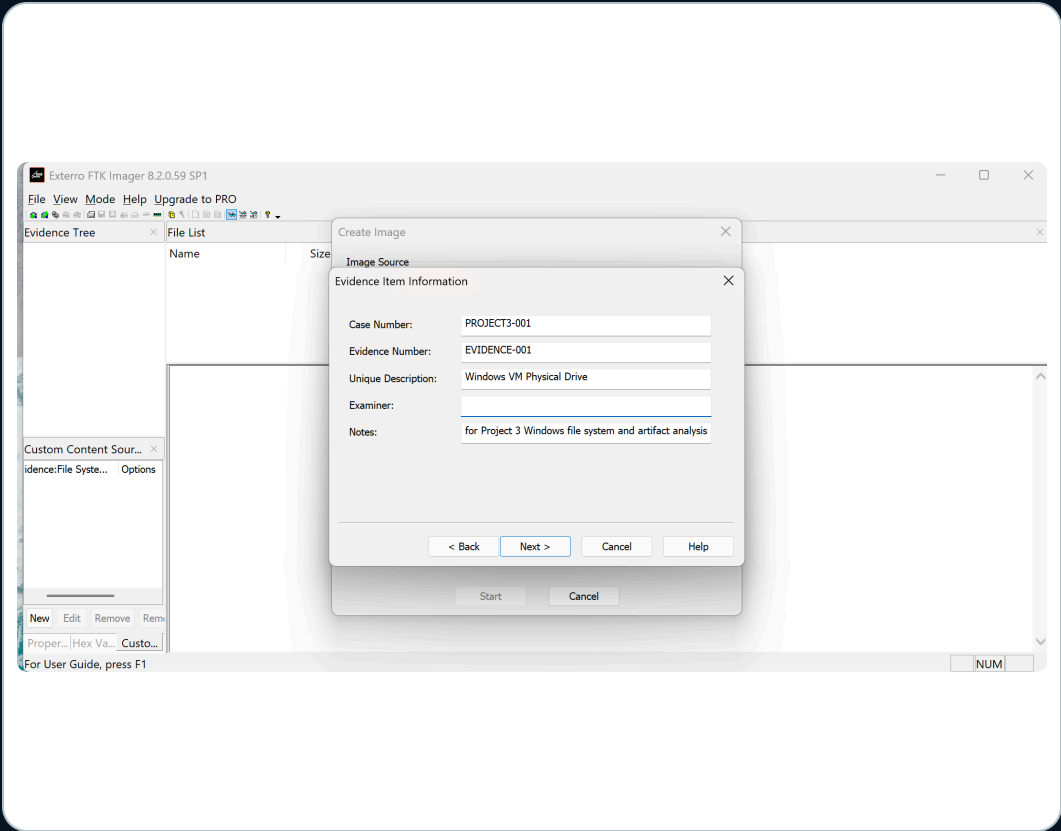
The tools complement one another: broad triage in Autopsy, with specialist parsers for deeper validation.

The source and destination were identified before imaging



SOURCE

Disk 0 contained the Windows system volume. Disk 1 was prepared as Evidence_storage (E:).



METADATA

Case PROJECT3-001 and evidence item EVIDENCE-001 documented the acquisition context.

Hash verification confirmed the E01 image was unchanged

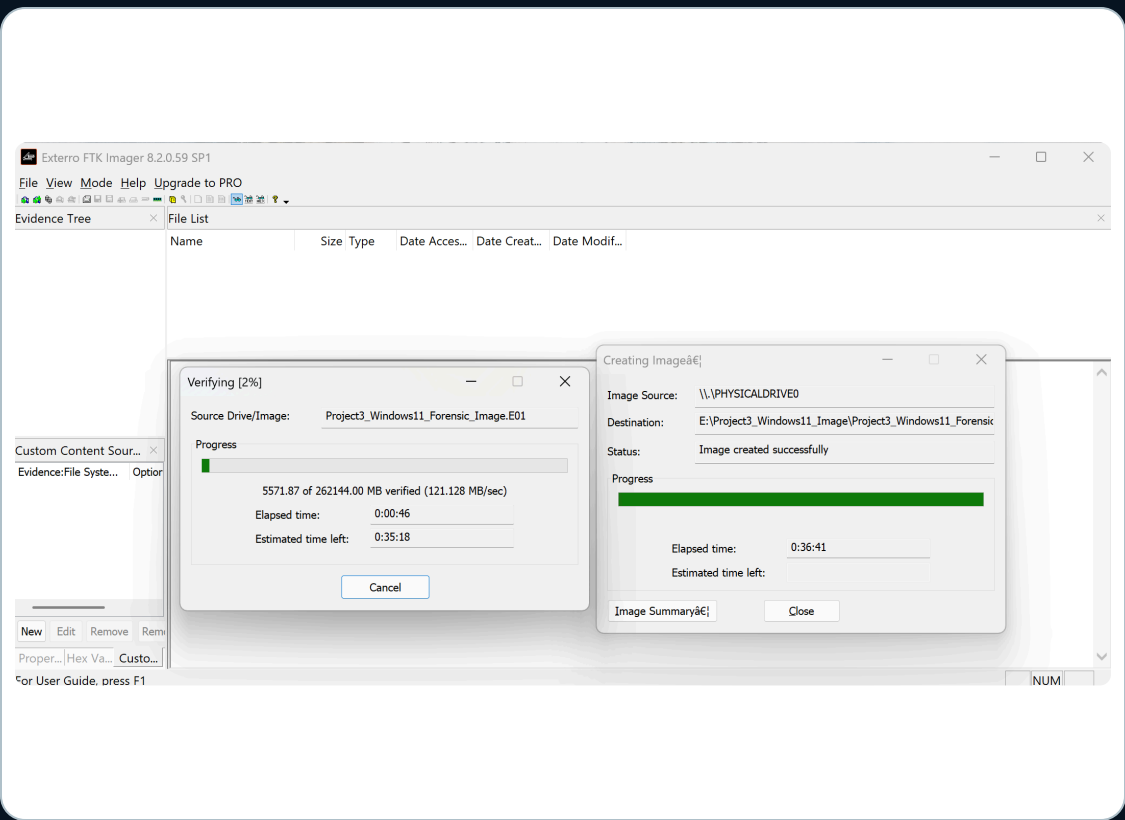
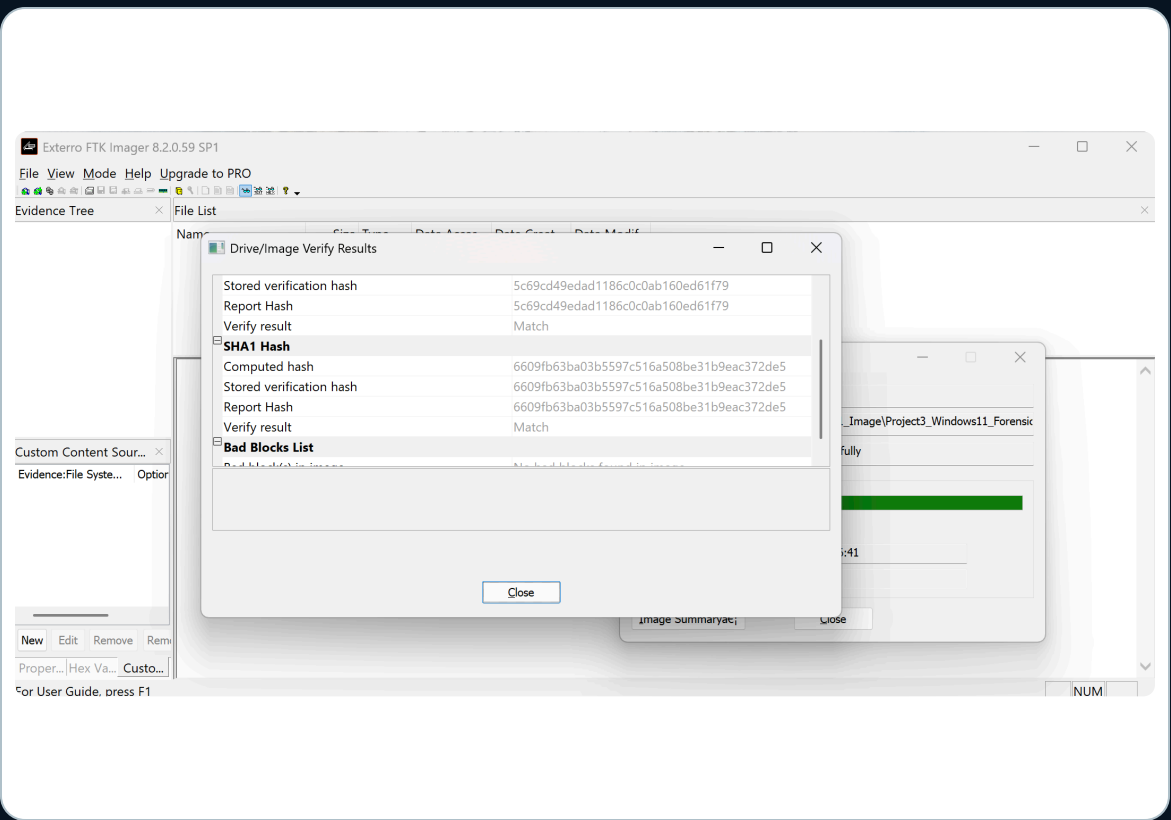


Image created successfully · 36:41 elapsed



MD5 match · SHA-1 match

Autopsy was configured for targeted artefact extraction

GeolocationTimelineDiscoveryGenerate ReportClose Case

Information

Case Information

Case Name:Windows11_Forensics_Project3

Base Directory:C:\Users\rileyworth3\Documents\Autopsy Cases

Case Type:☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:
C:\Users\rileyworth3\Documents\Autopsy Cases\Windows11_Forensics_Project3

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Finish

Cancel

CASE

GeolocationTimelineDiscoveryGenerate ReportClose Case

Data Source

Select Data Source

Path:E:\Project3_Windows11_Image\Project3_Windows11_Forensic_Image.E01

☐ Ignore orphan files in FAT file systems

Time zone:(GMT+0:00) Europe/London

Sector size:Auto Detect

Bitlocker Password (optional):

Hash Values (optional):
MD5:
SHA-1:
SHA-256:

NOTE: These values will not be validated when the data source is added.

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DATA SOURCE + TIME ZONE

GeolocationTimelineDiscoveryGenerate ReportClose Case

Data Source

Configure Ingest

Run ingest modules on:
All Files, Directories, and Unallocated Space

☒ Recent Activity

☐ Hash Lookup

☒ File Type Identification

☒ Extension Mismatch Detector

☒ Embedded File Extractor

☐ Picture Analyzer

☒ Keyword Search

☐ Email Parser

☐ Encryption Detection

☒ Interesting Files Identifier

☐ Central Repository

☐ PhotoRec Carver

☐ Virtual Machine Extractor

☒ Cloud Storage

☒ Cryptocurren

☒ Encryption Ph

☒ Privacy Progr

☒ Remote Mor

Select interesting fi

Identifies interesting

Select All

Deselect All

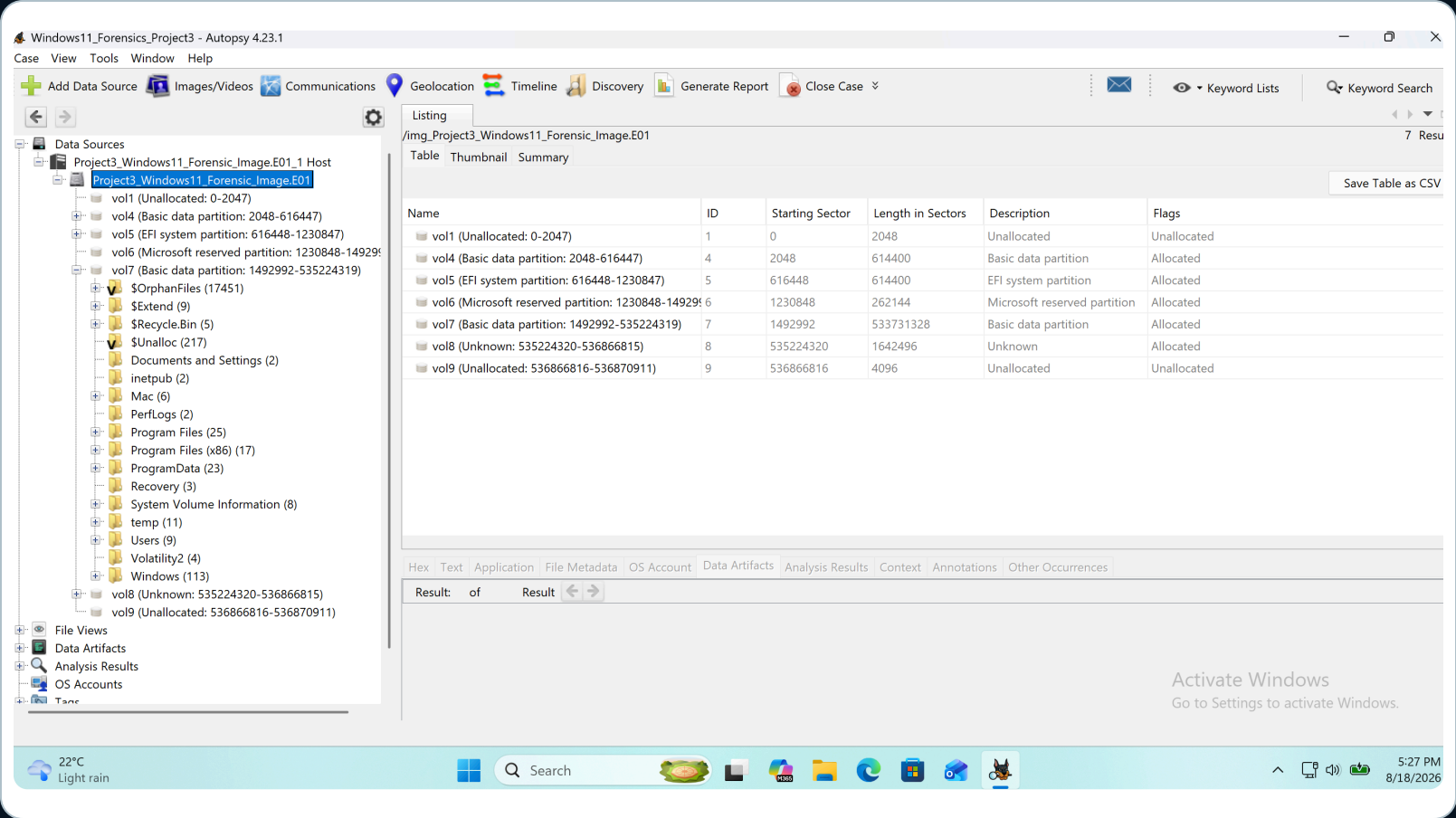
History

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INGEST MODULES

The image contained seven addressable volume entries



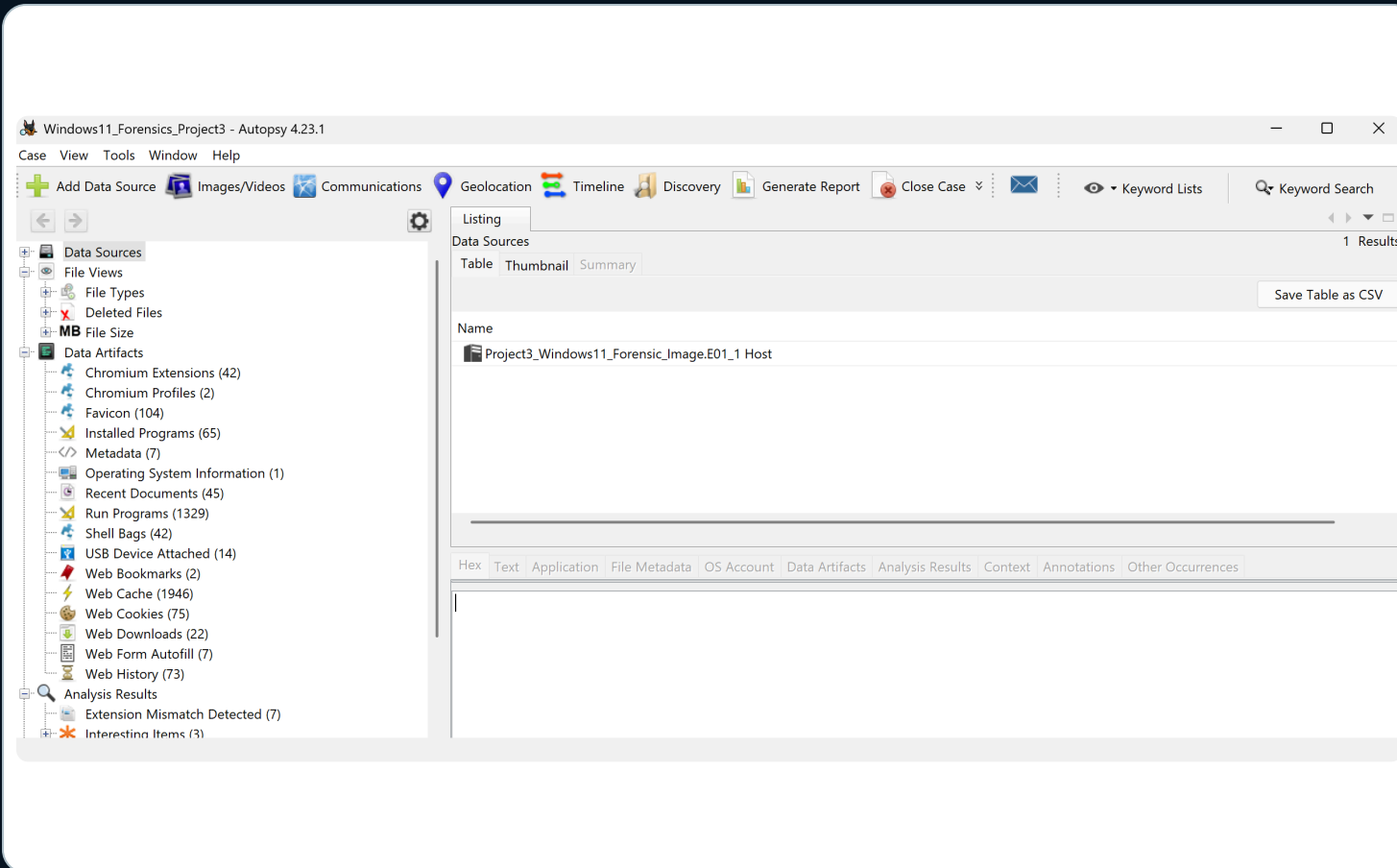
STRUCTURE

Autopsy identified the EFI system partition, Microsoft reserved partition, the main basic-data partition, an unknown partition and unallocated regions.

PRIMARY ANALYSIS AREA

Volume 7 contained the Windows directories, user profiles, Recycle Bin, orphan files and unallocated records used for artefact review.

Autopsy exposed a broad set of user-activity artefacts



1,329 Run Programs

1,946 Web Cache

73 Web History

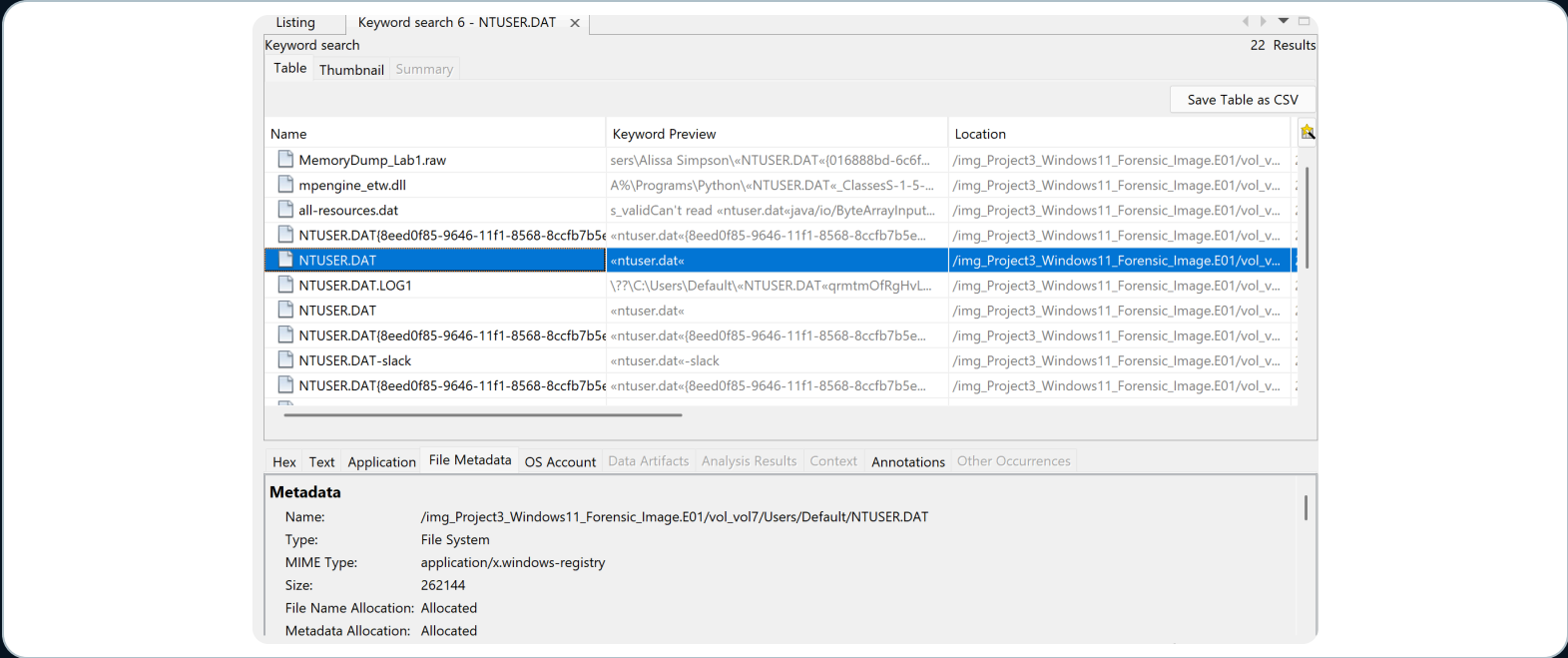
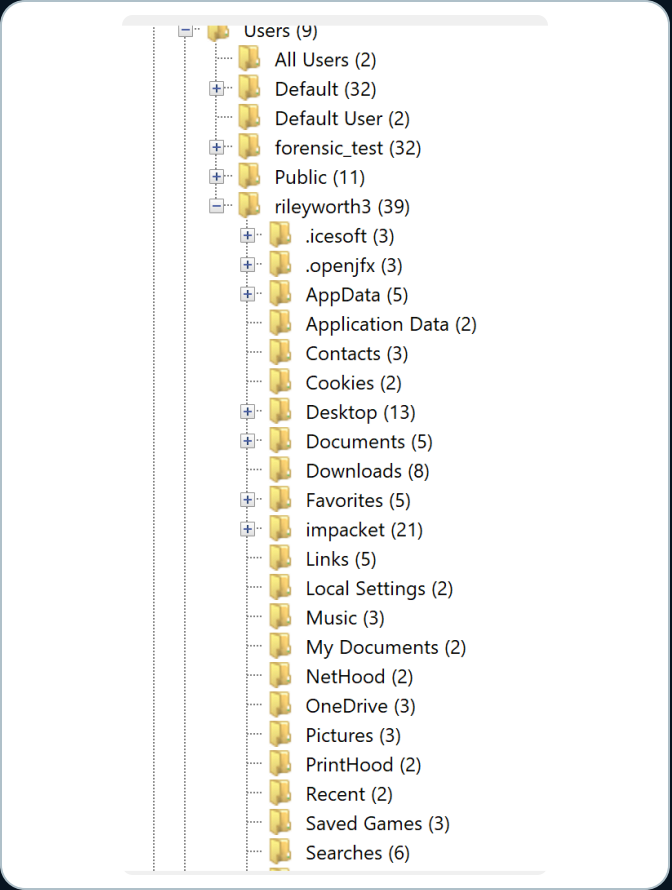
65 Installed Programs

45 Recent Documents

42 Shell Bags

14 USB attachments

User profiles established the ownership context for later artefacts



PROFILE STRUCTURE

The image contained forensic_test, rileyworth3, Default and Public profiles. The rileyworth3 tree exposed standard folders alongside tools and working locations.

REGISTRY SCOPE

The NTUSER.DAT search returned 22 results, including allocated hives, transaction logs and slack entries. These artefacts provide per-user configuration and activity context.

Shell Bags reconstructed visible folder paths

Data Sources

Project3_Windows11_Forensic_Image.E01_1 Host

Project3_Windows11_Forensic_Image.E01

File Views

Data Artifacts

Chromium Extensions (42)

Chromium Profiles (2)

Favicon (104)

Installed Programs (65)

Metadata (7)

Operating System Information (1)

Recent Documents (45)

Run Programs (1329)

Shell Bags (42)

USB Device Attached (14)

Web Bookmarks (2)

Web Cache (1946)

Web Cookies (75)

Web Downloads (22)

Web Form Autofill (7)

Web History (73)

Analysis Results

Shell Bags

TableThumbnailSummary

Save Table as CSV

Source Name	S	C	O	Path	Key	Data Source
UsrClass.dat				My Computer	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				My Computer\C:\	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				My Computer\C:\Mac	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				My Computer\C:\Mac\Home	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				My Computer\C:\Mac\Home\Documents	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				{f874310e-b6b7-47dc-bc84-b9e6b38f5903}	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\C:\Users\rileyworth3\Desktop	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\C:\Users\rileyworth3\Desktop\u001a\Desktop	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\Desktop	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\desktop	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\desktop\u001a\Desktop	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01
UsrClass.dat				Explorer\Downloads	Local Settings\Software\Microsoft\Windows\Shell\B...	Project3_Windows11_Forensic_Image.E01

42 RECORDS

Autopsy parsed Shell Bag entries from UsrClass.dat, preserving folder-navigation evidence even when the folders themselves may no longer be present.

OBSERVED LOCATIONS

Visible paths included C:\Mac\Home\Documents, C:\Users\rileyworth3\Desktop and Downloads-related Explorer entries.

INTERPRETATION

These records support prior folder visibility or navigation; they should be correlated with timestamps and other artefacts before claiming file access.

System metadata established the host context — with one caveat

TableThumbnailSummary

Save Table as CSV

Source Name	S	C	O	Name	Program Name	Processor Architecture	Temporary Files Directory	Path
Project3_Windows11_Forensic_Image.E01				RILEYWORTH5FC	Windows 10 Pro	ARM64	%SystemRoot%\TEMP	C:\V

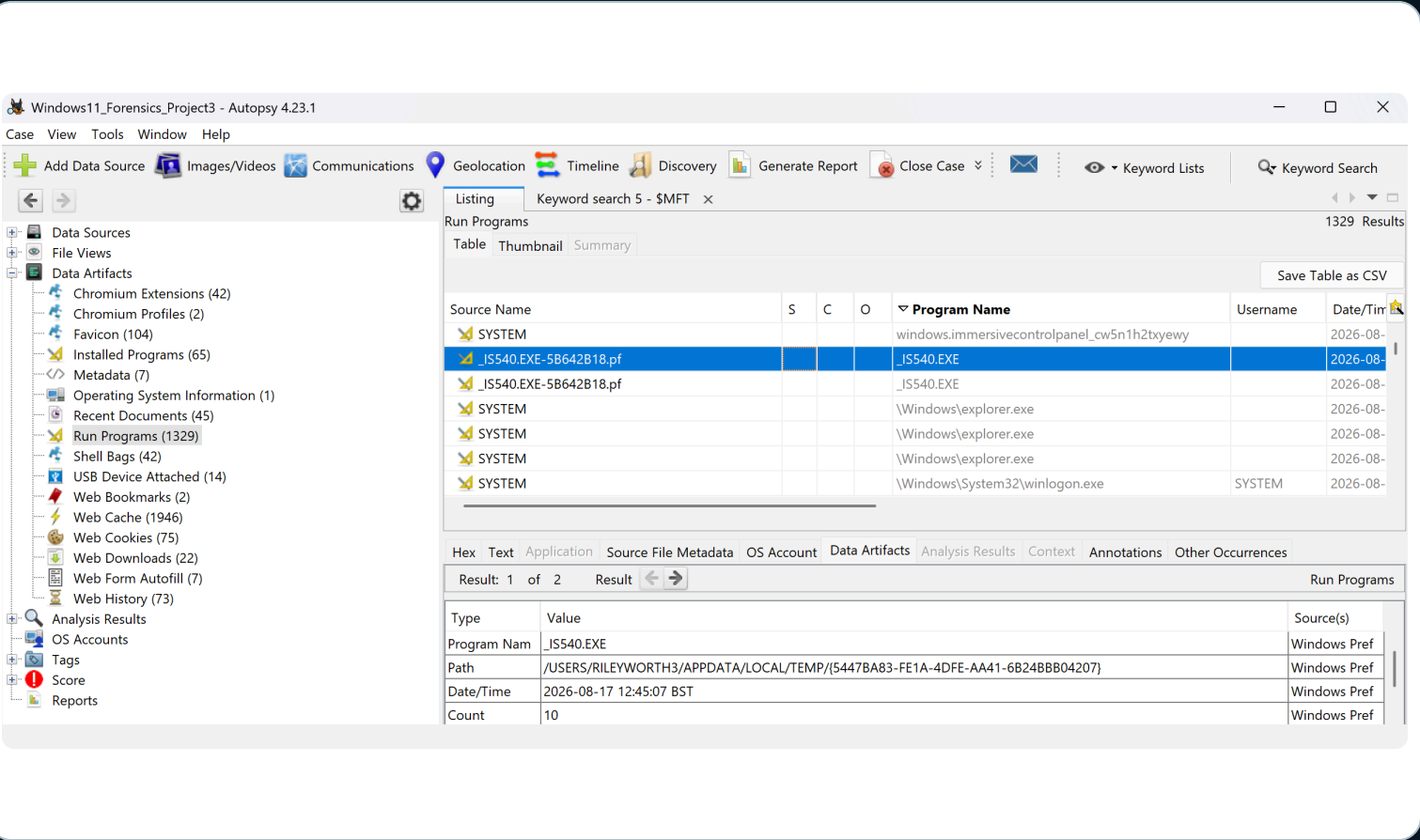
HexTextApplicationSource File MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotationsOther Occurrences

Result: 1 of 1Result<=>

Operating System Information

Type	Value	Source(s)
Name	RILEYWORTH5FC	Recent Activity
Program Name	Windows 10 Pro	Recent Activity
Processor Archit	ARM64	Recent Activity
Temporary Files	%SystemRoot%\TEMP	Recent Activity

Prefetch evidence supported repeated program execution



_IS540.EXE

Execution count

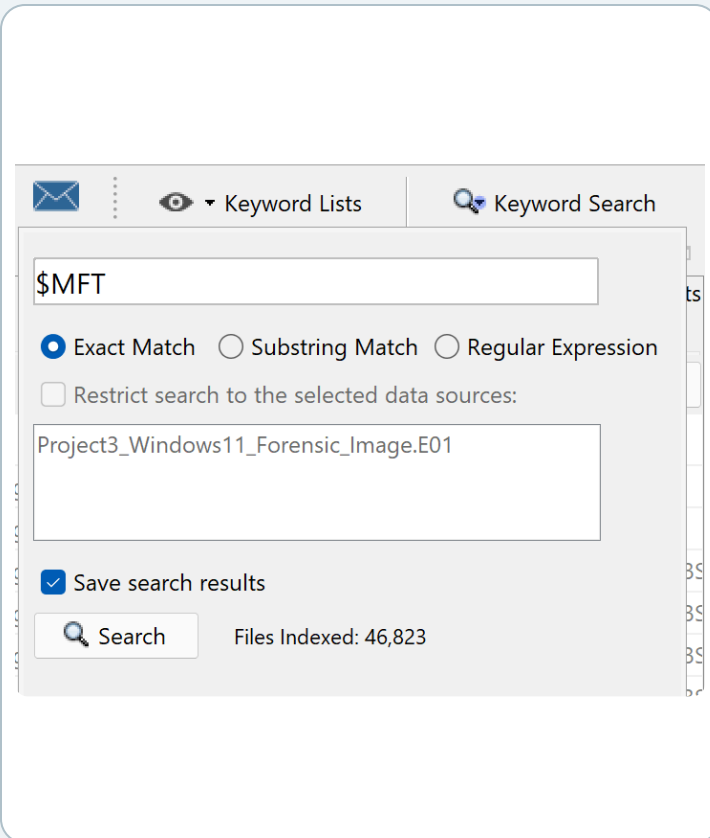
10

Recorded timestamp

17 Aug 2026
12:45:07 BST

Prefetch supports execution, but the timestamp and run count must be interpreted within Windows Prefetch behaviour and the wider timeline.

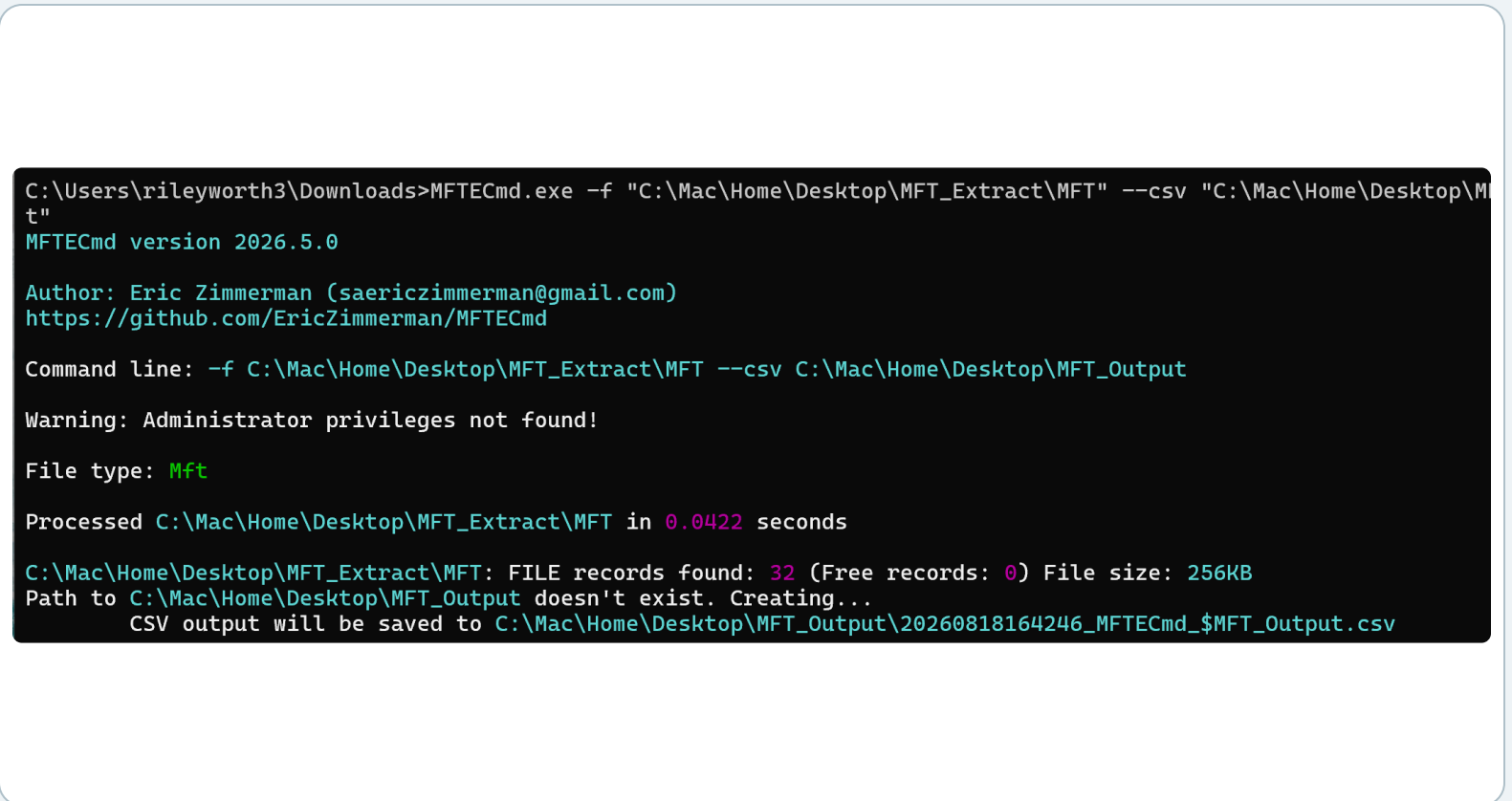
MFTECmd converted an extracted MFT sample into reviewable CSV



The screenshot shows the MFTECmd application window. At the top, there are tabs for 'Keyword Lists' and 'Keyword Search'. The 'Keyword Search' tab is active. Below it, there is a search term input field containing '\$MFT'. Underneath the search term, there are three radio buttons: 'Exact Match' (selected), 'Substring Match', and 'Regular Expression'. Below these, there is a checkbox labeled 'Restrict search to the selected data sources:' which is unchecked. Underneath the checkbox, there is a text box containing 'Project3_Windows11_Forensic_Image.E01'. At the bottom left, there is a checkbox labeled 'Save search results' which is checked. To the right of this checkbox, there is a 'Search' button. Below the 'Search' button, it says 'Files Indexed: 46,823'.

LOCATE

An \$MFT artefact was extracted from the forensic image for targeted parsing



```
C:\Users\rileyworth3\Downloads>MFTECmd.exe -f "C:\Mac\Home\Desktop\MFT_Extract\MFT" --csv "C:\Mac\Home\Desktop\MFT_Output\MFT.csv"
MFTECmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/MFTECmd

Command line: -f C:\Mac\Home\Desktop\MFT_Extract\MFT --csv C:\Mac\Home\Desktop\MFT_Output\MFT.csv

Warning: Administrator privileges not found!

File type: Mft

Processed C:\Mac\Home\Desktop\MFT_Extract\MFT in 0.0422 seconds

C:\Mac\Home\Desktop\MFT_Extract\MFT: FILE records found: 32 (Free records: 0) File size: 256KB
Path to C:\Mac\Home\Desktop\MFT_Output\MFT.csv doesn't exist. Creating...
CSV output will be saved to C:\Mac\Home\Desktop\MFT_Output\20260818164246_MFTECmd_$MFT_Output.csv
```

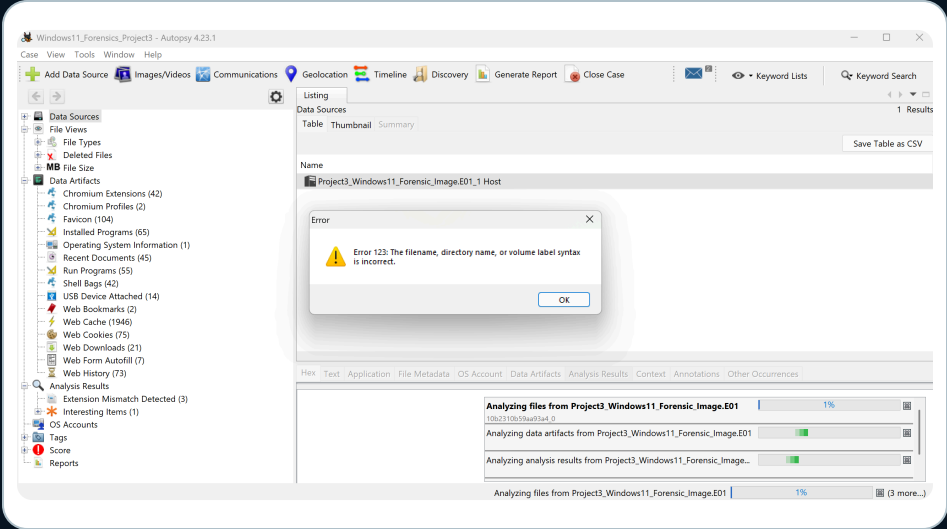
PARSE

MFTECmd 2026.5.0 parsed the 256 KB sample, identified 32 records and created a timestamped CSV.

The timeline also captured a tool crash and a path error

Faulting application name: WFA.exe, version: 2.10.0.0, time stamp: 0x6151e11f
 Faulting module name: unknown, version: 0.0.0.0, time stamp: 0x00000000
 Exception code: 0xc0000409
 Fault offset: 0x76f8c660
 Faulting process id: 0x26D0
 Faulting application start time: 0x1DD2E755B10BDAC
 Faulting application path: C:\Users\rileyworth3\Downloads\WFA.exe
 Faulting module path: unknown
 Report Id: d7bf81c5-c063-4a36-939a-fd01012db561
 Faulting package full name:
 Faulting package-relative application ID:

Log Name:	Application		
Source:	Application Error	Logged:	8/17/2026 7:22:27 PM
Event ID:	1000	Task Category:	Application Crashing Events
Level:	Error	Keywords:	
User:	RILEYWORTH5FC\rileyworth	Computer:	RILEYWORTH5FC
OpCode:	Info		



EVENT EVIDENCE

Event ID 1000 recorded WFA.exe crashing from the Downloads directory at 19:22:27 on 17 August 2026.

Lesson: preserve error evidence, correct paths and rerun with validated syntax.

The evidence supports a coherent, repeatable investigation

What the investigation established

Integrity: the E01 image passed MD5 and SHA-1 verification.

System context: host, architecture and partition structure were identified.

Activity evidence: Autopsy exposed profile, registry, Shell Bag, execution, web and USB artefacts.

Targeted validation: Prefetch evidence and a limited extracted MFT sample were examined with specialist tooling.

Lessons carried forward

Record case identifiers and acquisition settings before imaging.

Verify hashes before analysis begins.

Corroborate tool-reported operating-system labels.

Treat automated artefact counts as leads, not conclusions.

Document errors and rerun using validated paths and privileges.

Result: a defensible portfolio that links each conclusion to visible forensic evidence.